

SCALABILITY & FUTURE PLAN

Date	29-09-2026
Team ID	SWTID-2026-6684
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	API Rate Limits & Latency Reliance on synchronous API calls to Gemini models for both text scripts and panel image generation.	High concurrency causes delayed comic panel rendering, potential timeouts, and rate limit errors (HTTP 429) during peak user usage.	High
2	In-Memory / Local Storage for Assets Generated panel images and story metadata are stored locally or in temporary cache.	Restricts server scaling (stateless instance deployment), risks data loss on server restarts, and limits user gallery history.	High
3	Single-Threaded Rendering Pipeline Sequential generation of panel text scripts, visual descriptions, and image rendering.	Slower total turn-around time per multi-panel comic book (e.g., a 6-panel comic takes up to 30-45 seconds to generate).	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single server web deployment handling limited concurrent sessions.	Implement a stateless backend architecture with load balancers (AWS ALB / GCP Cloud Load Balancing) and auto-scaling container groups (Kubernetes/ECS).
2	Data Storage	Local file system storage for comic panels and temporary session state.	Migrate asset storage to AWS S3 / Cloud Storage with CDN (CloudFront) distribution, and use MongoDB / PostgreSQL for persistent story scripts and user profiles.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Synchronous generation pipeline causing rendering delays.	Introduce asynchronous background job processing using Redis Celery / Cloud Tasks for multi-panel generation, enabling parallel panel rendering and WebSockets for real-time progress updates.
4	Security	Basic client-side API usage with minimal authentication and rate limiting.	Implement JWT-based OAuth2 authentication, backend API key encryption key-vaults, fine-grained RBAC, and rate-limiting middleware (Nginx / Cloudflare WAF).

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Character Consistency Engine & Style Fine-Tuning Implement LoRA / character embeddings with Gemini Imagen context tuning to keep characters consistent across all panels.	Month 1 - Month 2	Significantly increases comic visual quality, narrative coherence, and user satisfaction.
Phase 3	Interactive Panel Editor & Audio Dubbing Drag-and-drop speech bubble customizer, dynamic layout templates, and AI voice-over narration generation for animated motion comics.	Month 3 - Month 4	Transforms static comics into rich multimedia stories, expanding content reach to video and audio formats.
Phase 4	Community Hub & Monetization Marketplace Publishing marketplace, social sharing, collaborative comic creation, and creator monetization subscriptions.	Month 5 - Month 6	Drives organic platform growth, user engagement, network effects, and recurring revenue generation.